

# Group 39 – Report

## 1. Titles

**Module title:** Software Projects with Agile Techniques

**CRN:** 34129

**Level:** 5

**Assessment title:** Assignment Two (HackCamp)

**Project link:** <http://columnist-carnival.poseidon.salford.ac.uk/HackCamp/index.php>

## 2. Executive Summary

This report documents the design, development, and evaluation of our group project website, “Find A Placement”. It is a responsive web app concept where it would match students with a placement based on their relevant skills and courses taken. The aim of the app was to address the struggle students have finding a suitable placement opportunity or to encourage them to take a year in industry by improving the way placements are advertised and matched through a skill-based system.

The solution allows students to manually upload a CV or manually select skills based on the SFIA v8 framework. The employer would post a placement with certain requirements; these requirements are being compared to the student's skill set, and then depending on the “weight” the algorithm gives, the placement is then matched with the student.

The system implements an MVC architecture, keeping the user data, interface, and control of the web app separate. Despite the challenges with scalability in this architecture, the team was able to deliver a functional proof of concept within the time restraints given.

Overall, the team was able to produce a working prototype that can demonstrate the core aspects of the application and provides a strong foundation for future development, despite the incomplete features.

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## 4. Introduction

Find A Placement is a responsive web application concept for the British Computing Society (BCS) which aims to address this issue by improving how students are matched to placement opportunities. The application allows employers to post placements and match students to optimal placements via skill-based matching. Students can input their skills through uploading a CV or using a dedicated dashboard. An algorithm is then used to compare student skills against placement requirements, helping to promote fairness by ensuring that all students are assessed fairly and provided with equal opportunities.

Moreover, it can allow for the possibility of career staff utilising its features by spotting placements that may be overlooked and supporting students through the placement process.

The purpose of this report is to cover the development process of the team's solution for the BCS. Each section outlines the scope of the assignment, the key requirements from the client brief, and the foundations of the team's development process. The report also describes how the team applied the Scrum framework during the project, including sprint planning and backlog refinement.

Furthermore, it explains what the team was expected to produce, how the development approach was organised, and how each stage of work contributed to the final application. It establishes the basis for the design, implementation, and testing activities, and provides a structured evaluation of the overall project's outcome.

## 5. Development Process

The team made sure the project followed the Scrum framework for developing the 3 sprints. Each sprint was only started after considering the sprint planning, justify the backlog, daily stand-up meetings, a customer demonstration and a sprint retrospective. Meeting records and supporting evidence are provided in the appendix.

The tools that the team decided on for this project are shown in Figure 1.7, with the tools name, justification and for what part of the project section.

The details of the distribution of assigned tasks and estimated time of completion are covered in each sprint under 5.2.1, 5.3.1 and 5.4.1 respectively.

The group would have about 2 stand-up meetings every day at the start and end of the day shown under 5.2.2, 5.3.2 and 5.4.2. The work was between 10:00am to 17:00pm. These meetings helped talk about project related issues, note progress and assign tasks. These meeting streamlined the project and was beneficial.

The sprint burn-down chart in sections 5.2.3, 5.3.3 and 5.4.3, covers the total remaining estimated time for each sprint. The line chart illustrates how much work remains at the start of each day. Showcasing whether the team is progressing ahead or behind the timeframe.

### Research

James did research on Notifications to find the requirements, what needed implementing as well as benefits and drawbacks, so the application can notify users about changes on the site. The research showed that it would require external resources and levels of security, such as needing the users to grant permission for notifications.

So, for the user benefits the symbol for notifications need to be recognisable, and before any information is sent had to be both explicitly availed at the users' permission and only include relevant and necessary notifications related to the user.

After the group listed tools and finalised the selection on 03/12/2024, James briefly detailed the tools use, benefits and drawbacks. The team got the basic understanding of the tools and if it was suited for frontend or backend. This allowed the team to select tools quickly for the given tasks.

Research was also done on GradCracker, LinkedIn and Indeed by James to understand the design layout of each website job page and deduce what was necessary for the project application to compete, while demonstrating any unique qualities.

Jacob did research on Existing Algorithms to understand what was needed to create an algorithm that can fetch skills from CVs and match them to placement skill requirements. It was found that artificial intelligence was used on the LinkedIn platform. It scans the user profile to fetch selective data to predict compatibility between the user and the job posting.

Research was conducted on GradCracker, LinkedIn and Indeed by Jacob and Jonathan to help the frontend visualise what the application will look. Mainly focusing on the initial view of the navigation bars, home and login pages. Noting any features that made each site unique or more beneficial.

Jonathan and James were responsible for researching the benefits of placements, to add content for the target audience which are Students and Employers. The goal is to increase awareness of the benefits and

opportunities placements offer for both parties. The goal is to have a dedicated page, containing the details with links from other pages and accessible in the footer. Users will be more willing to use a site which purpose is clear.

## 5.1 Product backlog for entire project

### Epic 1 – Design and Implement Home & Article page

1. Design the project in a development tool such as Figma for reference of UI.
2. Home page with logo, hero section, about us, out partners, overview of how SFIA matching works and footer.
3. Article page with navigation section. Navigation contains student benefits of pursuing a year in industry, explanation of SFIA skills and how to apply.

### Epic 2 – Login and Registration

1. Secure login system based on user role.
2. Verify and hash password against database.
3. Registration form for all user, password type check. (e.g must have a number)

### Epic 3 – Student Profile and Skill Management

1. Create a student profile page that displays personal information and skills.
2. Set the requirement that CVs are supposed to be in PDF format.
3. Fetch skills from the uploaded CV based on the SFIA framework.
4. Allow the Student to delete or edit fetched skills list.
5. Store the Student's skill data into the database.

### Epic 4 – Employer Profile and Placement Management

1. Set up an employer dashboard.
2. Create a placement posting form.
3. Validate placement information.
4. Store the placement data into the database.
5. Show all the hosted placements.
6. Allow the employers to update the placements.

### Epic 5 – Career Staff

1. Create staff profile to allow university career staff to assist students.
2. Allow staff to view students from the associated university.
3. Allow the Staff to view and explore for placement recommendations-based the associated student's profile.

### Epic 6 – Admin

1. Allow for viewing, management and checks of user accounts when logged in.
2. Remove or suspend malicious users
3. Monitor placement posting to reduce fake or malicious posts.

## 5.2 Sprint One (2 Dec 2025 – 4 Dec 2025)

### 5.2.1 Sprint backlog, work item estimates, work item selection by group members

| Spring Backlog Tasks                 | Assigned To | Estimate Time of Completion |
|--------------------------------------|-------------|-----------------------------|
| 1. Gather requirements for home page | Sujon       | 1 Hour                      |
| 2. Design home page in Figma         | Sujon       | 3 Hour                      |

|                                                                    |                 |         |
|--------------------------------------------------------------------|-----------------|---------|
| 3. Design system architecture                                      | Jacob           | 2 Hour  |
| 4. Design database                                                 | Joe and Henry   | 1 Day   |
| 5. Gather requirement for article page                             | Jon and James   | 2 Hour  |
| 6. Design article page                                             | James and Raees | 1 Hour  |
| 7. Implement navigation                                            | Jacob and Sujon | 2 Hour  |
| 8. Link all the navigation items                                   | James           | 1 Hour  |
| 9. Hero section explaining the platform and showing analytics data | James and Sujon | 3 Hour  |
| 10. Write the about us section for home page                       | James           | 1 Hour  |
| 11. Write why students should do placements for article page       | Jonathan        | 1 Hours |
| 12. Write why employers should offer placements for article page   | Jonathan        | 1 Hour  |
| 13. Implement student registration                                 | Jonathan        | 1 Hour  |
| 14. Implement student login                                        | Jonathan        | 1 Hour  |
| 15. Design student dashboard                                       | Jonathan        | 1 Hour  |
| 16. Implement employer registration                                | Jacob           | 1 Hour  |
| 17. Implement employer login                                       | Jacob           | 1 Hour  |
| 18. Implement employer dashboard                                   | Jacob           | 1 Hour  |
| 19. Implement employer profile                                     | Jacob           | 1 Hour  |
| 20. Implement admin login                                          | Joe             | 1 Hour  |
| 21. Implement admin dashboard                                      | Joe             | 1 Hour  |
| 22. Implement career staff login                                   | Henry           | 1 Hour  |
| 23. Implement career staff dashboard                               | Henry           | 1 Hour  |
| 24. Design placement form                                          | Raees           | 2 Hour  |
| 25. Implement placement description                                | Raees           | 1 Hour  |
| 26. Implement user type login                                      | Jacob           | 1 Hour  |
| 27. Validate passwords and hash credentials                        | Jacob           | 1 Hour  |

### 5.2.2 Daily stand-up meeting

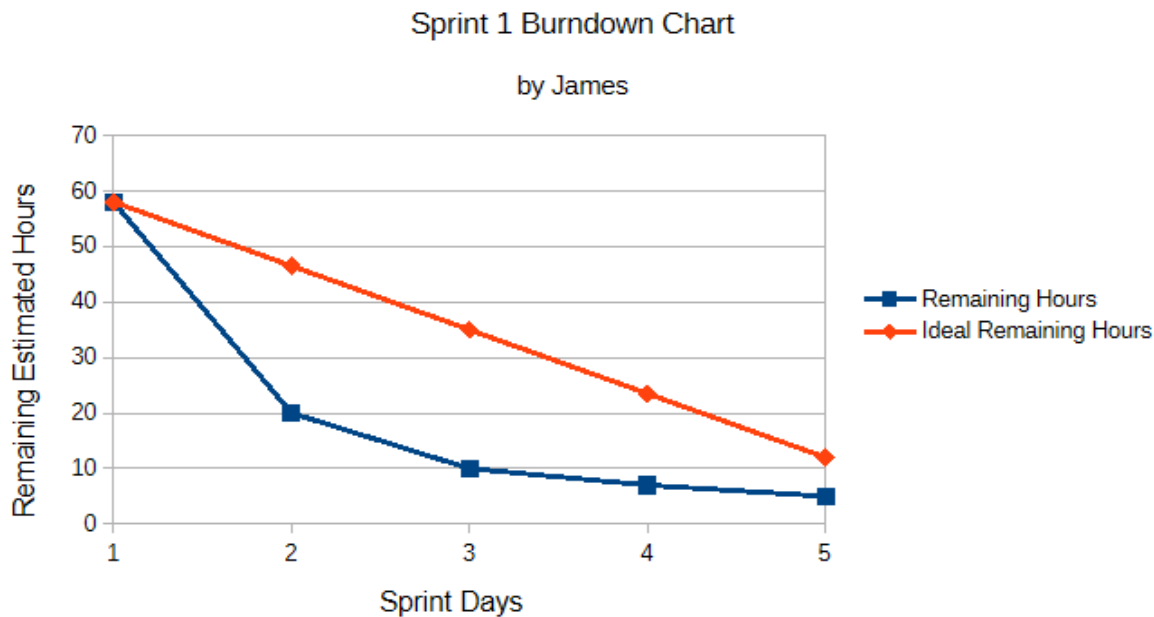
| Meeting Numbers | Date & Time                          | Location     | Attendees                            | Discussion                                                                                                                                                                                                                     | Decisions                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|--------------------------------------|--------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1               | 2 <sup>nd</sup> Dec 2025<br>10:00 AM | SEE building | Sujon, Henry, James, Joe, Jon, Jacob | <ul style="list-style-type: none"> <li>- Discussed who should be scrum master.</li> <li>- Discussed what tech stack should we use to implement the project.</li> <li>- Discussed the forms of group communications.</li> </ul> | <ul style="list-style-type: none"> <li>- Group decided Sujon should be scrum master.</li> <li>- We decided to use JavaScript, Bootstrap, HTML5, CSS, PHP, SQLite, and Git for source code management.</li> <li>- We decided to use WhatsApp for communication and updates.</li> </ul>                                                                                                                                                           |
| 2               | 2:30 PM                              |              |                                      | <ul style="list-style-type: none"> <li>- Discussed the progress of work accomplished.</li> <li>- Discussed the work planned for the day.</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>- Team decided to review and analyse the project brief.</li> <li>- Research into languages, tools and competitors to justify both use and application, research into benefits of placements.</li> </ul>                                                                                                                                                                                                  |
| 3               | 5:00 PM                              |              |                                      | <ul style="list-style-type: none"> <li>- Discussed work plan before next meeting</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>- Team decided to work to complete research tasks individually.</li> </ul>                                                                                                                                                                                                                                                                                                                               |
| 4               | 3 <sup>rd</sup> Dec 2025<br>10:00 AM | SEE building | Sujon, Henry, James, Joe, Jon, Jacob | <ul style="list-style-type: none"> <li>- Discussed the progress work accomplished between meetings.</li> <li>- Discussed the days agenda.</li> <li>- Discussed any issues preventing progress.</li> </ul>                      | <ul style="list-style-type: none"> <li>- The team decided to discuss the facts, tools and competitors, so to make sure that research was completed.</li> <li>- The team decided to work on designing the homepage and login, fact explanation for article page as well as set up the necessary classes and tables.</li> <li>- The team noted that project requirements lacked clarity and decided that client contact was necessary.</li> </ul> |
| 5               | 5:00 PM                              | SEE building |                                      | <ul style="list-style-type: none"> <li>- Discussed the in-between meeting work plan.</li> </ul>                                                                                                                                | <ul style="list-style-type: none"> <li>- Team decided to start the implementation of the homepage, login page, creation of the basic navigation bar and send an email about a meeting time.</li> </ul>                                                                                                                                                                                                                                          |

|   |                                          |              |                                             |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|------------------------------------------|--------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | 4 <sup>th</sup> Dec 2025<br><br>10:00 AM | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Catch up what was accomplished between meetings.</li> <li>- Discussed the work planned for the day.</li> </ul>                                                    | <ul style="list-style-type: none"> <li>- Team decided to work on the design for the article page,</li> <li>- design and implementation for student dashboard elements, such as account activity page and account detail editing,</li> <li>- work on admin dashboard design and implementation,</li> <li>- design and implementation of placements page.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7 | 5:00 PM                                  | SEE building |                                             | <ul style="list-style-type: none"> <li>- Discussed work planned prior to next meeting.</li> </ul>                                                                                                          | <p>Team decided that two members are to:</p> <ul style="list-style-type: none"> <li>- Research pdf readers/parsers in preparation for skill matching algorithm.</li> <li>- Work on front-end debugging / styling for admin dashboard and other sections.</li> <li>- Implementing functions for admin dashboard, bug-fixing with placement, research and starting of employer dashboard.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8 | 5 <sup>th</sup> Dec 2025<br><br>10:00 AM | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Catch up of work accomplished between meetings.</li> <li>- Discussed work planned for the day.</li> <li>- Discussed any issues preventing progression.</li> </ul> | <p>Team confirmed that the:</p> <ul style="list-style-type: none"> <li>- Basic pdf reader was created.</li> <li>- Front-end for admin dashboard has improved.</li> <li>- Back-end for admin dashboard now improved.</li> <li>- Placement pages bugs are fixed.</li> </ul> <p>Team decided that the:</p> <ul style="list-style-type: none"> <li>- skill matching algorithm is to be created,</li> <li>- skills list and synonyms is to be created,</li> <li>- placements page getting more filters and better front-end,</li> <li>- employer dashboard being improved,</li> <li>- creating placement descriptions,</li> <li>- admin controller and models being improved,</li> <li>- admin dashboard work continued,</li> <li>- function being created to compare placement skills and student skills</li> </ul> |
| 9 | 5:00 PM                                  | SEE building |                                             | <ul style="list-style-type: none"> <li>- Discussed what will be completed over the weekend.</li> </ul>                                                                                                     | <p>Team decided that:</p> <ul style="list-style-type: none"> <li>- Users should be able to manually select skills for their accounts</li> <li>- The skill matching algorithm is to be finished</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|  |  |  |  |  |                                                                                                                                                                                                                                         |
|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  |  | <ul style="list-style-type: none"> <li>- Fix bugs</li> <li>- Fix the employer dashboard bug issues</li> <li>- Work on formatting, design, placement page bug fixes and</li> <li>- To have the automated testing to be set up</li> </ul> |
|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Due to unforeseen personal mitigating circumstances, a team member was ill during the first 2 days of hackcamp, the team discussed that regular updates would be sent through WhatsApp and Jira on what happened during each meeting. Then full debriefing the team member when they returned.**

### 5.2.3 Sprint burn-down chart



### 5.2.4 Customer demonstration meeting

Due to complications with setting up an appointment with the client such as time conflicts, the project manager opted to collectively ask questions to understand the brief better (Figure 1.1) and justify the Product Backlog items as shown in Figure 1.2. Figure 1.3. Furthermore, it allowed the group to better prioritize tasks based on client and implementation requirements.

### 5.2.5 Sprint retrospective

#### What went well was:

- That the tasks were within the planned timeframe.
- Starting to implement the foundation of the website.
- The weekend was used to refine any issues, add data and complete remaining. features.

#### What went wrong was:

- That the team did not set up a customer demonstration meeting, due to not requesting a meeting in a professional setting.

#### Actions for next sprint is:

- Set a client meeting that meets professional standards.
- Create a documentation of the progress for the report.

## 5.3 Sprint Two (5 Dec 2025 – 8 Dec 2025)

### 5.3.1 Sprint backlog, work item estimates, work item selection by group members

| Spring Backlog Tasks                                                  | Assigned To     | Estimate Time of Completion |
|-----------------------------------------------------------------------|-----------------|-----------------------------|
| 1. Write our mission section for home page                            | James and Sujon | 1 Hour                      |
| 2. Write the overview of how matching works for home page             | Sujon and James | 1 Hour                      |
| 3. Write a documentation for the report                               | James           | 5 days                      |
| 4. Write our partners section for home page                           | Sujon           | 1 Hour                      |
| 5. Gather research insights for NCUB study for article page           | Jonathan        | 1 Hour                      |
| 6. Write How SFIA skills match improve employability for article page | Jonathan        | 1 Hour                      |
| 7. Design the Profile management for student system                   | Jonathan        | 3 Hour                      |

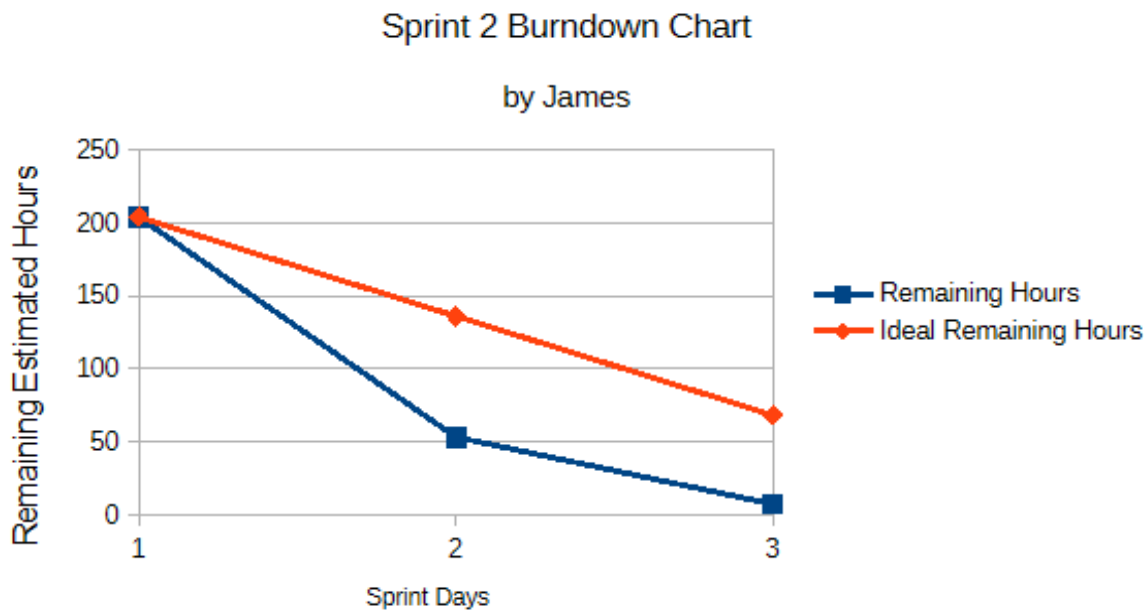
|                                                                                    |                 |        |
|------------------------------------------------------------------------------------|-----------------|--------|
| 8. Design the SFIA v8 skills input for student system                              | Jonathan        | 3 Hour |
| 9. Implement the CV upload in PDF format                                           | Jonathan        | 3 Hour |
| 10. Design the view for matched placements according to SFIA v8 for student system | Jonathan        | 4 Hour |
| 11. Design placement for employer system                                           | Jacob           | 3 Hour |
| 12. Implement the editing or deletion of placements for employer system            | Jacob           | 4 Hour |
| 13. Implement the management of placement listings for employer system             | Jacob           | 5 Hour |
| 14. Implement setting required SFIA skills for employer system                     | Jacob           | 4 Hour |
| 15. Implement required SFIA skills for placement adding system                     | Raees           | 5 Hour |
| 16. Implement the salary input for placement adding system                         | Raees and James | 2 Hour |
| 17. Implement the location input for placement adding system                       | Raees and James | 2 Hour |
| 18. Implement the start and end date input for placement adding system             | Raees and James | 2 Hour |
| 19. Implement the placement detail page for placement adding system                | James and Raees | 2 Hour |
| 20. Implement student to placement matching via SFIA v8 for matching engine        | Raees and Jon   | 5 Hour |
| 21. Implement employer to student matching for matching engine                     | Raees and Jon   | 5 Hour |
| 22. Design the management of students for admin system                             | Joe             | 5 Hour |
| 23. Design the management of employers for admin system                            | Joe             | 2 Hour |
| 24. Design the management of placements for admin system                           | Joe             | 2 Hour |
| 25. Design the management of career staff for admin system                         | Joe             | 2 Hour |
| 26. Design the view of all students for career staff system                        | Henry           | 3 Hour |
| 27. Design the view of all placements for career staff system                      | Henry           | 3 Hour |
| 28. Design the view matching results for career staff system                       | Henry           | 3 Hour |
| 29. Design placement page                                                          | Sujon           | 1 Hour |
| 30. Design login pages                                                             | Sujon           | 1 Hour |
| 31. Design login page                                                              | Sujon           | 1 Hour |
| 32. Design registration page                                                       | Sujon           | 1 Hour |
| 33. Design employer page                                                           | Sujon           | 1 Hour |
| 34. Design admin page                                                              | Sujon           | 1 Hour |
| 35. Design Career staff page                                                       | Sujon           | 1 Hour |

### 5.3.2 Daily stand-up meeting

| Meeting Numbers | Date & Time                          | Location     | Attendees                                   | Discussion                                                                                                                                              | Decisions                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------|--------------------------------------|--------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10              | 8 <sup>th</sup> Dec 2025<br>10:00 Am | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Catch up what was accomplished over the weekend.</li> <li>- Discussed the work planned for the day.</li> </ul> | <p>Team confirmed that:</p> <ul style="list-style-type: none"> <li>- The matching algorithm still needs improvement,</li> <li>- Data entry into placement and employer field in database is complete,</li> <li>- Fixed bugs,</li> <li>- Employer dashboard still has issues</li> <li>- Team decided to start implementing features</li> <li>- The team decided to start write sections for each page</li> <li>- Team decided to assign James to document the progress</li> </ul> |
| 11              | 5:00 PM                              | SEE building |                                             | - Discussed work planned prior to next meeting.                                                                                                         | - Team decides to continue individual tasks                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 12              | 9 <sup>th</sup> Dec 2025<br>10:00 Am | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Catch up what was accomplished between meetings.</li> <li>- Discussed the work planned for the day.</li> </ul> | <ul style="list-style-type: none"> <li>- The team confirmed that the pages designs have been completed</li> <li>- Information entry is complete</li> <li>- The team decided to implement login and sign in features</li> <li>- The team decided to work designing the matching engine for student and employer</li> </ul>                                                                                                                                                        |
| 13              | 5:00 PM                              |              |                                             | - Discussed work planned prior to next meeting.                                                                                                         | - The team decided to complete the additional features                                                                                                                                                                                                                                                                                                                                                                                                                           |

|    |                                       |                 |                                                   |                                                                                                       |                                                                                                                                                                                                                                                                     |
|----|---------------------------------------|-----------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                       |                 |                                                   |                                                                                                       | - The team decided to work on work new client tasks                                                                                                                                                                                                                 |
| 14 | 10 <sup>th</sup> Dec 2025<br>10:00 Am | SEE<br>building | Sujon, Henry,<br>James, Joe, Jon,<br>Jacob, Raees | - Catch up what was accomplished<br>between meetings.<br>- Discussed the work planned for the<br>day. | - Addition of necessary features are complete<br><br>- All pages are a single theme<br><br>- Noted that a select accounts do not function<br><br>- The team decided to merge all the branches in<br>GitHub for testing<br><br>- Design the view layout of each page |
| 15 | 5:00 PM                               |                 |                                                   | - Discussed work planned prior to next<br>meeting.                                                    | - Fix bugs<br><br>- The team decided to work on adding any remain<br>features before code freeze                                                                                                                                                                    |

### 5.3.3 Sprint burn-down chart



### 5.3.4 Customer demonstration meeting

The client meeting was from 10:30 to 11:01 on 09/12/2025. The team summarized the first sprint and showcase the current progress of the project. Due to mitigating circumstances, one of the clients was not in attendance. The tasks that were discussed are the administration links, location, user accounts and password resets, documentation and if the site was up to General Data Protection Regulations (GDPR).

The discussion of GDPR, user accounts, and passwords are related to if solutions were in place to deal with cyber security attacks, false information and accounts, while not reducing user experience due to negligence by administrators who did not misuse the system.

### 5.3.5 Sprint retrospective

#### What went well was:

- The website's foundation is completed within the planned timeframe.
- It was very easy knowing what to implement for each page because of the research done before the sprint started.
- Example data was created during the weekend, so fetching data from the database can be used to implement the design of the website.
- This sprint a meeting with the client was properly set to start a customer demonstration meeting.

#### What went wrong was:

- The team had multiple merge conflicts because of the database, this slowed down the progress as data was not properly uploaded.
- Though the team chose tools that each member could use or explain the functionality, both the differing competency levels of each member and the team collaboration introduced unfamiliar situations.

#### Action for next sprint is:

- Start testing the application.
- Set the final client meeting to demonstrate the progress made after the last meeting.
- Confirm changes to the database with a team member that understands GitHub to reduce changes of creating merge conflicts.

## 5.4 Sprint Three (9 Dec 2025 – 12 Dec 2025)

### 5.4.1 Sprint backlog, work item estimates, work item selection by group members

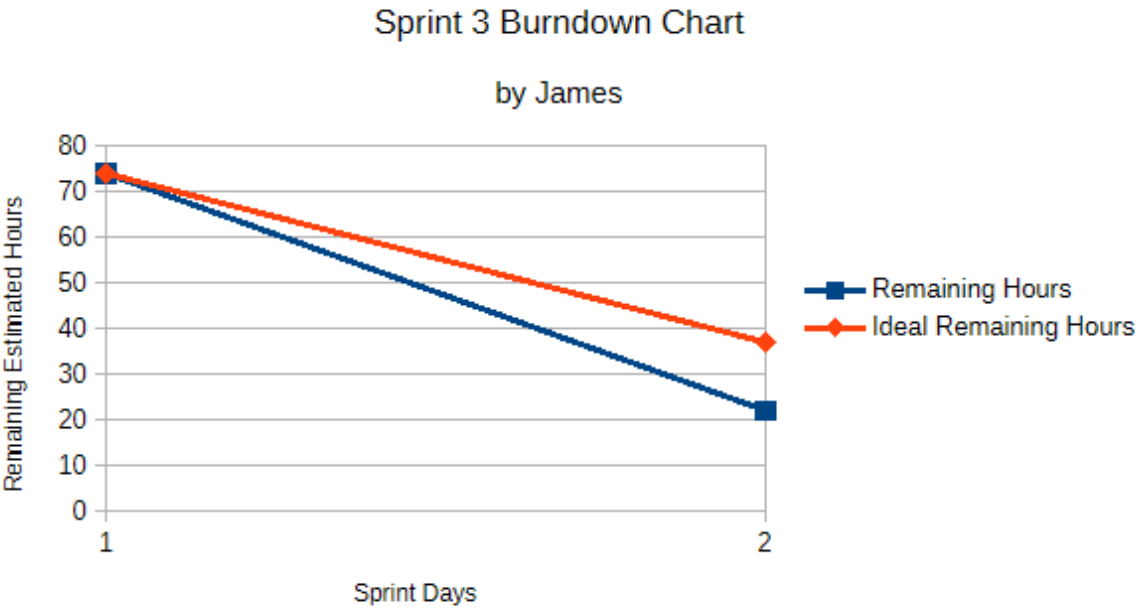
| Spring Backlog Tasks                                                              | Assigned To     | Estimate Time of Completion |
|-----------------------------------------------------------------------------------|-----------------|-----------------------------|
| Design the footer                                                                 | Sujon           | 4 Hour                      |
| Write placement process explanation for article page                              | Jonathan        | 1 Hour                      |
| Implement the profile editing for student system                                  | Jonathan        | 3 Hour                      |
| Implement email notifications for student system                                  | Jonathan        | 4 Hour                      |
| Design the view for matching students for employer system                         | Jacob           | 4 Hour                      |
| Implement email notifications for employer system                                 | Jacob           | 4 Hour                      |
| Secure authentication for all users                                               | Jacob and Sujon | 4 Hour                      |
| Implement placement search and filters for placement adding system                | Raees           | 5 Hour                      |
| Implement automation of match scoring for matching engine                         | Raees           | 5 Hour                      |
| Implement match notification email for matching engine                            | Raees           | 5 Hour                      |
| Implement moderation tools for admin system                                       | Joe             | 4 Hour                      |
| Implement the reset passwords for admin system                                    | Joe             | 3 Hour                      |
| Implement the adjustments of matching setting for admin system                    | Joe             | 3 Hour                      |
| Implement assist function with student profiles or skills for career staff system | Henry           | 4 Hour                      |
| Implement access CVs for career staff system                                      | Henry           | 5 Hour                      |
| Implement a more responsive design for the frontend                               | Sujon           | 6 Hours                     |
| Implement CV file storage                                                         | Sujon           | 4 Hour                      |
| Review application to check GDPR compliance                                       | Sujon           | 6 Hour                      |

### 5.4.2 Daily stand-up meeting

| Meeting Numbers | Date & Time                           | Location     | Attendees                                   | Discussion                                                                                                                                              | Decisions                                                                                                                                                                                                                                                                                                                         |
|-----------------|---------------------------------------|--------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16              | 11 <sup>th</sup> Dec 2025<br>10:00 Am | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Catch up what was accomplished between meetings.</li> <li>- Discussed the work planned for the day.</li> </ul> | <ul style="list-style-type: none"> <li>- Implemented some of the additional features</li> <li>- The team decided to test the website for issues</li> <li>- The team decided to work on noting the issues for documentation</li> <li>- The team decided to focus on implementing security features after client meeting</li> </ul> |
| 17              | 5:00 PM                               | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Discussed work planned prior to next meeting.</li> </ul>                                                       | <ul style="list-style-type: none"> <li>- All unresolved issues were relayed for documentation</li> <li>- Work on security features</li> </ul>                                                                                                                                                                                     |
| 18              | 12 <sup>th</sup> Dec 2025<br>10:00 Am | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | <ul style="list-style-type: none"> <li>- Catch up what was accomplished between meetings.</li> <li>- Discussed the work planned for the day.</li> </ul> | <ul style="list-style-type: none"> <li>- Noted any security concerns</li> <li>- Noted unresolved bugs</li> </ul>                                                                                                                                                                                                                  |

|    |         |              |                                             |                                                        |                                                                                                                                                                                                                                       |
|----|---------|--------------|---------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |         |              |                                             |                                                        | - The team checked for any features that need implementing before code freeze                                                                                                                                                         |
| 19 | 5:00 PM | SEE building | Sujon, Henry, James, Joe, Jon, Jacob, Raees | - Discussed work to be done before the report deadline | - The team decided to review the code<br><br>- The team decided to split tasks into fixing the code and restructuring the documented progress into the report<br><br>- The team decided to distribute tasks to complete individually. |

5.4.3 Sprint burn-down chart



5.4.4 Customer demonstration meeting

The client meeting was from 10:30 to 11:00 on 11/12/2025. The team provided a detailed summary and demonstration of the product and the development at the last meeting. Due to mitigating circumstances, only the client was in attendance. The meeting started with questions on the security issue, what was done to make sure that career staff were required to gain permission to alter CVs. There were also questions about what could be done or should have done to make the product better. The overview by the client addressed the need to fix as much code before the freeze.

The pages that were worked on by each member in subgroups were also pointed out the need to be more cohesive about design and presentation across the website. Documentation was briefly reviewed, and the client stated that it was important to also document the intension of each task as well.

## 5.4.5 Sprint retrospective

### What went well was:

- The security issues or concerns have been addressed with the clients during the meeting.
- The client like the produced MVP and knows that documentation has been done.
- Has no additional issues with the page consistency and uniform theme.

### What went wrong was:

- There was that some bugs were found and caused issues that are hard to fix.
- The team did not have time to fully implement all the additional features.

## 6. Requirements Analysis and System design

### 6.1 User stories

There are five types of users of the system. They are

- Student
- Employer
- Career staff
- Admin
- Browsing user

Below is the list of actions each user type can perform.

#### Student:

- As a student, I want to register so I can apply and shortlist placement opportunities.
- As a student, I want to follow an employer so I can explore more placement opportunities for them later.
- As a student, I want to get automated email job alerts, so I know when there is a new placement opportunity.
- As a student, I want to delete my shortlisted job so I can remove jobs that I am not interested in anymore or that I have already applied for.
- As a student, I want to search keywords on the placement page so I can find placement that suits my skills without manual inspection.
- As a student, I want to add my CV so that I get matching placements with my criteria and auto-upload SFIA-matching skills to the system.
- As a student, I want to add new skills to my skill section so that I get an increase in placement matches.

#### Employer:

- As an employer, I want to register so that I can access website functionality to post jobs.
- As an employer, I want to post placement opportunities so I can hire interns.
- As an employer, I want to delete a placement so that I can remove opportunities that are no longer available.

#### Career staff:

- As university career staff, I want to register so I can view students from my university.
- As a career staff member, I want to view and explore placements so I can recommend them to my university students.

#### Admin:

- As an admin, I want to be available to log in so I can moderate the website and do security checks.

- As an admin, I want to manage user accounts such as student, employer, and career staff so that I can remove malicious accounts.
- As an admin, I want to view placements so I can deter fake and malicious posts.

Browsing user:

- As a browsing user, I want to view the placements listing so that I can explore opportunities without registering.
- As a browsing user, I want to view the home page so that I can learn about the website and how it functions.
- As a browsing user, I want to view the article section to read about how to apply and how SFIA matching works.

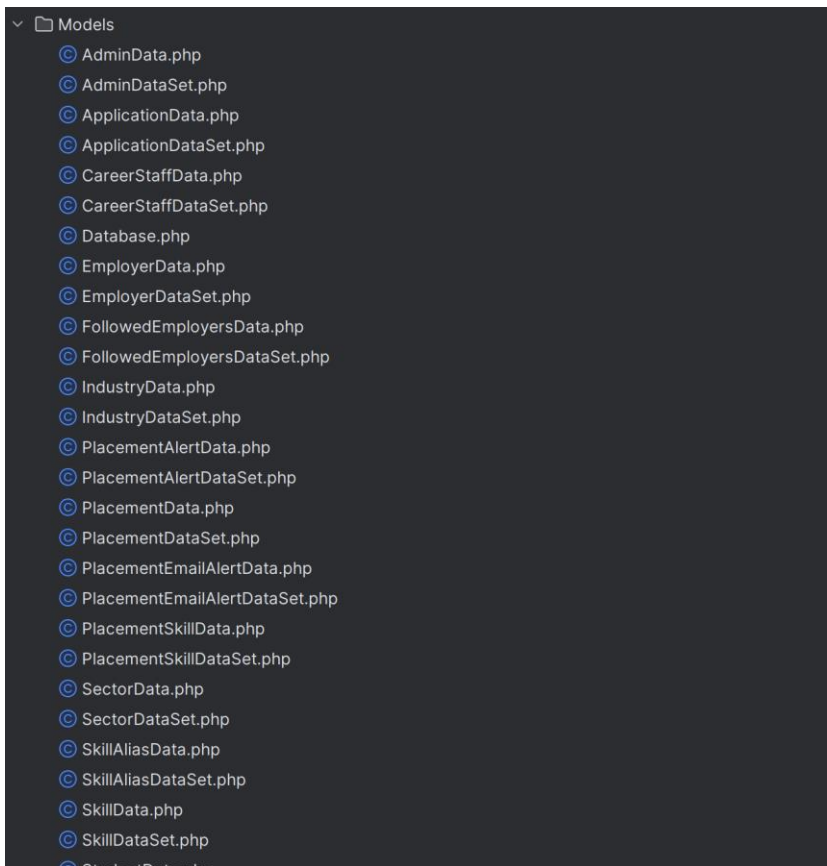
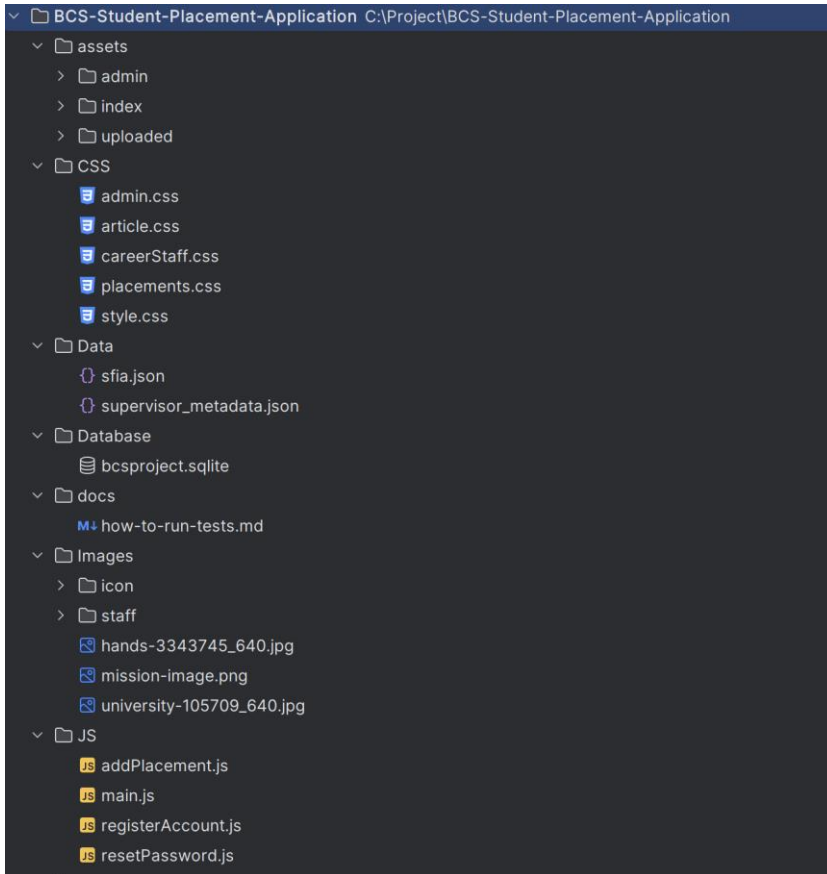
## 6.2 Overall system architecture

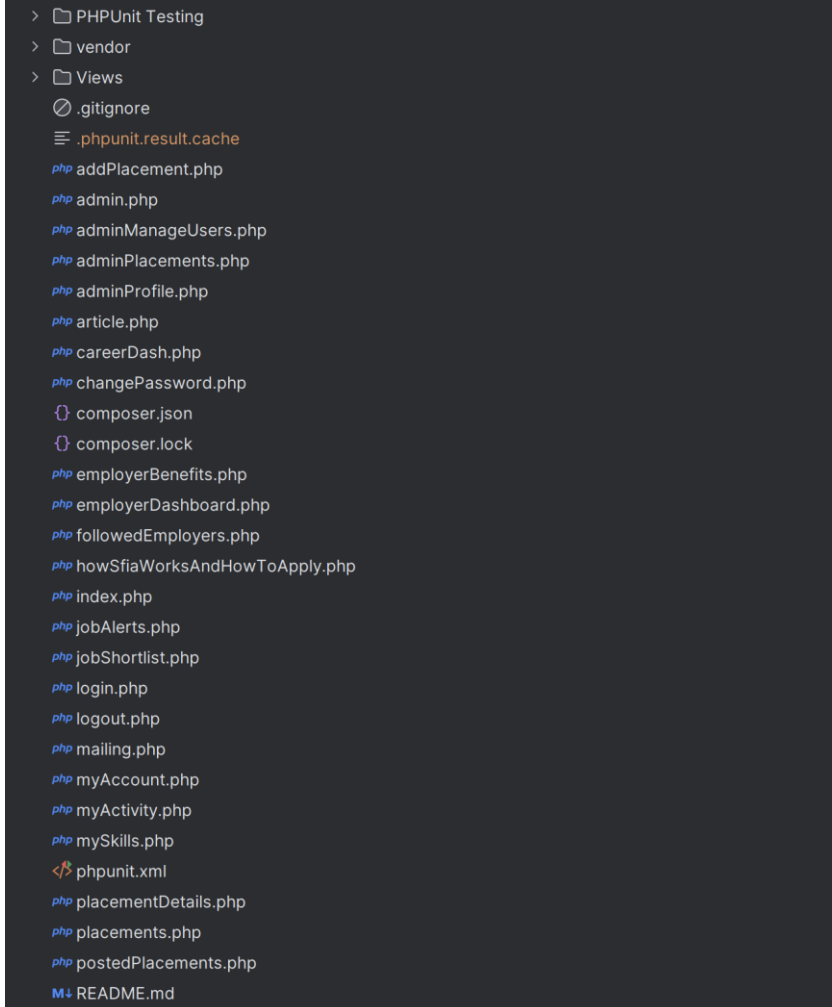
We have decided to use Model View Controller (MVC) architecture for this project as all team members were familiar with the concepts through previous university course work. We assumed the MVC pattern will be manageable and scalable.

But after sprint two, we have realized that it was not the best choice, because the root of the project was getting populated by many files and the sub directory system could have been more manageable if everything was feature based. For an example, the architecture “Feature Driven Modular Architecture” could have been a better choice.

Because our project has 5 different types of users and many features. If the system architecture was built in feature base model and managed in sub directory for each feature, it could have been more scalable in long run. This is because if we had to implement a new feature there will be no need for adding all view and model of different feature in the same directory as how MVC pattern does.

However, it was too late to scrap the process, and we have managed to finish the project successfully. We have learnt that in a big project environment, architecture of the system should be decided depending on the project needs, not by the level of familiarity possessed by the developers. We realized that we could have learn a new architecture easily and implement that for scalable solutions. Nevertheless, the MVC architecture of this project is shown in below figure.



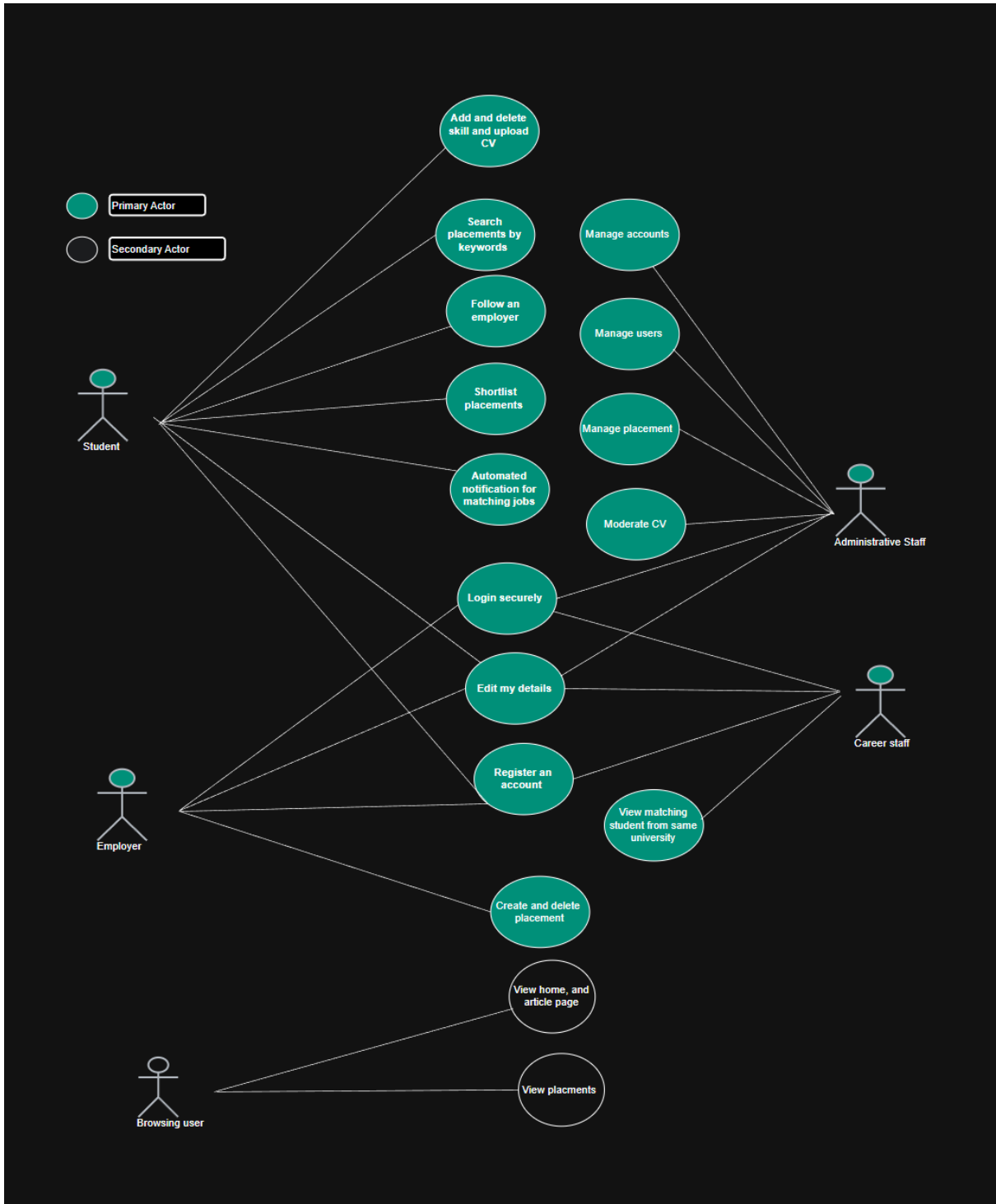


A screenshot of a file explorer window showing a project directory. The directory contains several folders and files. The folders are: > PHPUnit Testing, > vendor, and > Views. The files are: .gitignore, .phpunit.result.cache, addPlacement.php, admin.php, adminManageUsers.php, adminPlacements.php, adminProfile.php, article.php, careerDash.php, changePassword.php, composer.json, composer.lock, employerBenefits.php, employerDashboard.php, followedEmployers.php, howSfiaWorksAndHowToApply.php, index.php, jobAlerts.php, jobShortlist.php, login.php, logout.php, mailing.php, myAccount.php, myActivity.php, mySkills.php, phpunit.xml, placementDetails.php, placements.php, postedPlacements.php, and README.md. The files are listed in a single column, with some files having a small icon to the left of the name.

## 6.3 System designs

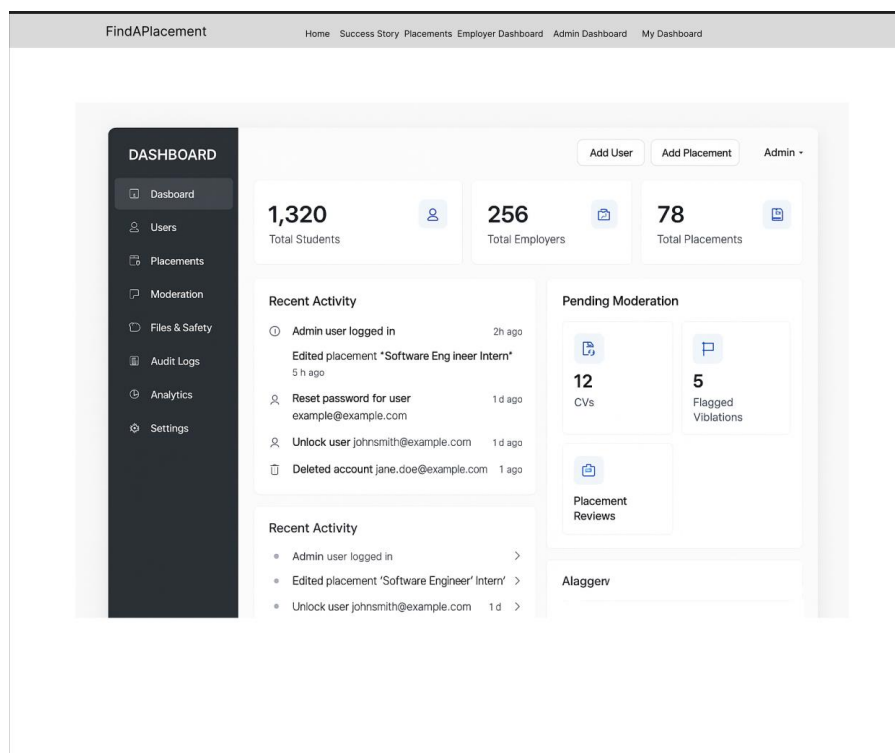
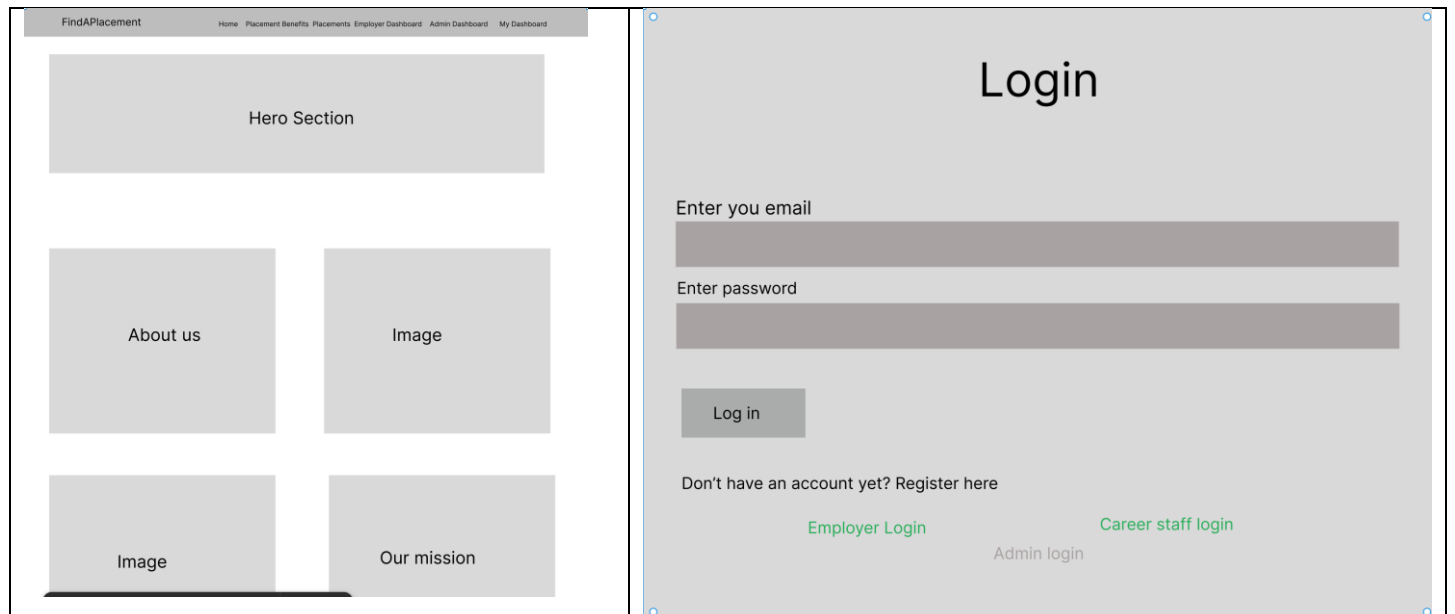
### 6.3.1 UML diagrams

The user stories are presented in the UML diagram below which helped the team understanding features concise and quickly.

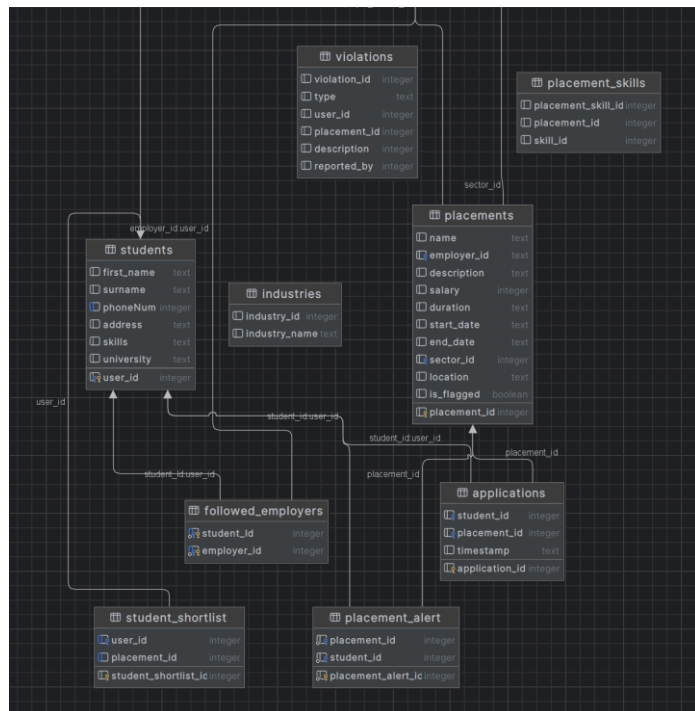
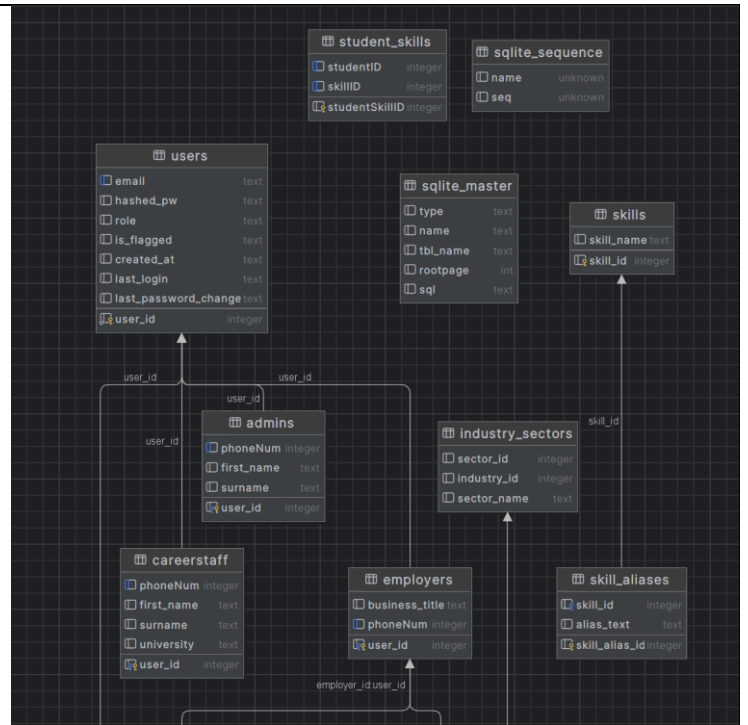
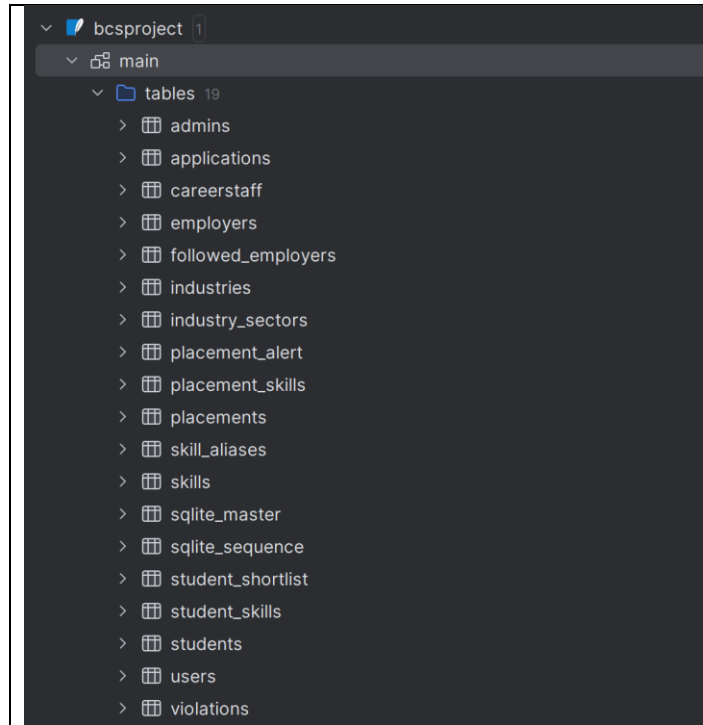


## 6.3.2 Website Design

The website design was done in Figma because it is widely use among software developer and free to use tool. Below is some proof of design. Please note that the implementation of the design has changed slightly during development based on needs. Here is few design demos amongst many:



### 6.3.3 Database design (SQLite)



## **7. Solution Implementation**

### **Site navigation**

Prior to a user's login the navigation bar can be used to move between the home page, placement benefits page, placements page, and login pages for each type of user. This navigation bar is accessible from all pages and is the main means of movement between all pages. Each page contains a footer at the bottom of the page. The footer is where the companies' contact details such as email address, phone number and address are displayed to the user.

Due to the limited time for the creation of the project, whilst we have set up links for the company's social media as well as links for the websites terms of service, cookie policy and privacy policy we have yet to create the related accounts or sample versions of these policies.

### **Home Page**

For our solution, we opted to minimise the number of features that appear on the homepage. Instead, we focused on using it to highlight the strengths of the website whilst giving insight into how the matching algorithm functions. Our homepage includes a live tracker for the number of placements, employers and users seeking placements to help encourage new users to utilise the website both for posting placements and finding them.

### **Article Page**

The brief specifies the need to explain to users the potential benefits of either hosting a placement as an employer or taking a placement year as a student. This page is viewable regardless of whether the user is logged in or not, which in turn encourages new users to use the website. This feature of the website is divided into three pages (student benefits, employer benefits, and SFIA skills and how to apply to placements) all accessible via a navigation bar at the top of each page.

Initially we planned to interview a student that had on a placement year to highlight the benefits of industry placements and make the website more appealing to students. However, we were unable to find a student that had been on a placement year that was available for an interview in the time remaining, so this feature had to be omitted.

### **Login**

To access the websites remaining features the user must login using their email and password. Each user account type has their own custom login portal that must be used to login to access their dashboard. The design of each portal is kept consistent so that users can login as quickly as possible.

Unfortunately, whilst our portals design does include options for password recovery and to keep the user logged in, we were unable to implement them as we opted to focus on more crucial features for website to function such as the algorithm and email notifications. We also initially considered setting up two factor authentication for the login process but lacked the time to do so after setting up the login process.

### **Placements**

As this is one of the most crucial pages of the website, this is also one of the pages with the most features. This page is where the user can search for and filter placements by a variety of factors. This includes the option to search for placements using a search bar, being able to set a minimum and maximum salary for placements, search for placements based upon how long they last as well as being able to filter placements based on what industry or sector that they are categorised as. Each placement found can initially be seen with the job title, description, employer, salary, duration and start date, however they can be selected to view more specific details like the location, sector and industry.

The placements page also serves as a way for students (when logged in) to follow employers and shortlist specific placements to view from their dashboard.

We were unable to give the user the ability to filter for placements across several industries or sectors in the same search due to time constraints. And whilst we were able to create several test versions of these placements, since these are not real placements users are unable to use the apply online button to apply for the placements currently on the website.

### **Student Account**

If a user logs in with a student account, then they are given access to several new pages as part of the student dashboard. These being My Activity, Job Shortlist, My Job Alerts, My Skills, Followed Employers and Account Settings. The My Activity page serves to summarise the accounts activity, so it contains features such as a live tracker for the number of followed employers, number of job alerts, number of placements shortlisted suggested employers to follow and a list of recently viewed placements. Whilst the My Activity page has the necessary area to view recently viewed placements, we were unable to get it functional within the given timeframe.

The My Jobs Shortlisted page is where any placements shortlisted by the user can be found and viewed in detail. This page works similarly to the placements page however placements on this page cannot be filtered or searched for. The followed employers page has a similar usage as it allows the user to view the posted placements for employers that they have followed but also gives the user the option to unfollow from employers.

My Skills page is how students can manually manage which SFIA skills are associated with the account. The user can select skills from a list and add them to the account or remove any skills accidentally associated with the account should they accidentally be added either by the user or by searching algorithm used to select SFIA skills from the users CV. These skills are used to match the user to placements which can then be recommended to them these placements can be seen on the My Job Alerts page each placement on this page can be selected to be viewed in increased detail.

Finally Account settings is a page used to manage the user's profile, this page allows the user to change account details such as email address, phone number, address, and profile picture. This page also allows the user to upload a CV so the website can pull SFIA skills from the PDF and add them to the account.

### **Employer Dashboard**

Much like the other dashboards, the employer Dashboard can only be viewed when logged in with the correct account. This dashboard has four pages that the user can interact with these being the Dashboard, Add Placement, View Placement and View potential Candidates pages.

The Dashboard page is where the user can manage their profile details such as email address and profile picture, as well as view information such as the number of posted placements and potential matches.

The add placement page is where employers fill in the relevant details for placements and upload them to the website to be viewed and applied to by students. This page requires information such as the placement's description, required skills (for automated matching), salary and location.

The view placements page is where employers can view and manage currently active placements. This page allows employers to delete currently posted placements if they are no longer available as well as amend any required SFIA skills for the placement. This page cannot currently be accessed by employers as we were unable to implement this feature in time.

## **Admin Dashboard**

To ensure the website can operate smoothly, we opted to make an account type not specified in the project brief, the admin. This account has access to its own dashboard as well as the homepage, articles page and placements page. The admin dashboard is broken down into 5 pages each with its own function. The dashboard, profile, manage users, placements and moderation pages. The dashboards main functions are to view the accounts recent activity as well as view any requests pending moderation. These requests can include CV's or placements and function to help the website avoid false placements or attempts at malicious behaviour.

The profile page is intentionally kept simpler as this page just serves to change the associated name and email address for the account should a transfer of role occur. The manage users page is how the admin can help moderate accounts on the website, this page allows the administrator to select specific accounts and view them, whether it be to delete an account that has been flagged or help the user with an issue regarding the account.

The placements page within the admin dashboard functions differently from the placements page the users use, this is because this how the admin manages placements. The admin can select specific placements and view their contents and from there decide how to proceed, whether it be leaving the placement viewable or deleting it if it violates the websites terms of service.

Given the time constraints, we were unable to implement all the features we initially set out to. The moderation page could not be implemented in time and so it is currently non-functional. In addition to this whilst we have most of the pages set up for the dashboard, unfortunately there are some bugs and issues in the Profile, manage Users and Placements pages that have left them unusable.

## **Career staff Dashboard**

As university career staff have limited usage for the website, we have only made one page for them. This being the Dashboard, this allows the user to change profile details as well as view any students associated with the university.

We initially considered allowing career staff to be able to recommend or shortlist specific placements to be sent to students associated with their university, however this was too large a task to be completed in the given timeframe.

## **Matching students and placements**

To match the students with placements that they are qualified for and notify them we had to break the issue down into two problems. The first being how we can tell if a student is a match for a placement, and the second being how we can notify the student about the matches.

To deal with the first issue we gave students two different means of adding skills to their accounts, they can either manually add SFIA skills to their accounts using the My skills page, and they have the option to upload a CV which is then searched through for SFIA skills which are then added to the account. These skills are compared against the skills that employers attach to their placements and as long as enough skills match then that placement is stored. Each placement will be compared against the student's skills, and the top ten placements are stored. This also means that in the future, the same placements will not be recommended to the student.

The second issue required us to use a mail server (ahasend) to email the matches to the student. This also meant that we needed to set up a company email account to maintain professional standards when contacting users. Unfortunately, this issue took longer to resolve than initially expected so our solution is that we are only able to email students recommended placements.

## 8. Details of the solution testing and evaluation

### 8.1 Acceptance Test Cases & Edge Cases

To make sure our application implemented the required functionality in the backend code, we defined acceptance criteria based on the User Stories. The table below demonstrates a few scenarios we tested against.

| Requirement            | Test Scenario                                      | Condition/Edge Case                                                                                                   | Expected Result                                                                          | Status |
|------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|
| Role-based Permissions | Verifying user permissions based on their ID.      | An edge case would be a student attempting to use admin privileges.                                                   | The system would return `false` when `isAdmin(USER ID)` is executed, denying the access. | Pass   |
| Handling Data          | Retrieving user details from the database.         | An edge case would be requesting a non-existent user ID (such as `0`).                                                | The system would return null and handle the error gracefully.                            | Pass   |
| Login Logic            | Identifying a user's role via their email address. | A condition would be ensuring that you can only login with a given role with the password associated with an account. | The system correctly assigns the user their given access level.                          | Pass   |

### 8.2 Unit tests

For testing, we focused on the backend logic (specifically the validity of models), to ensure that the method in which we handled data, as well as the user permissions, functioned correctly, before connecting them to the frontend. Since the development time was limited to 2 weeks, we used an automated unit testing suite rather than manual testing, which aided in development speed.

The testing framework we used was PHPUnit (version 12.5.4), to test the model correctness. For example, we explicitly tested the `AdminDataSet` model, to verify that non-existent IDs returned null in the PHP code and were handled accordingly in calling code. As modifying the live database during testing is unhygienic, an in-memory SQLite3 database that was seeded using copy of the database during production was used. The specific class was `TestDatabase`, which implemented the setup and teardown of the test env. Every test class then called the test database's setup and teardown functions in the test's own respective functions. This ensured that each test was not contaminated by the results of other tests, maintaining data hygiene.

### 8.3 Test results

The test suite covered every function of every model in our `Models/` folder. The focus was to ensure that every input gave us the resulted output, thoroughly checking every returned attribute, rather than a 'success'. This is because deterministic code creates an ideal production environment. The results of the test suite are as follows:

- **Tests Ran:** 50
- **Total Assertions:** 1,769
- **Execution Time:** 0.368 seconds
- **Memory Usage:** 18 MB

The large number of assertions made (which averaged roughly 35 per test) confirms that we are successfully retrieving the *correct* data that we expect our models to return from the database.

While the test suite did have a 100% pass rate, confirming the logic's correctness, the output *did* return a notice for every test, which is due to the fact that our database singleton is meant to be a global object, but we assign a new `TestDatabase` instance to it each time, which causes PHPUnit to trigger warnings on test isolation and

static state manipulation, despite the fact our code's hygiene is valid. However, we should resolve this in future work, as using a singleton, while convenient, makes composability more difficult.

While the backend code was covered via automated testing, the frontend, the views and controllers, were tested manually against the criteria, to ensure that the design was responsive on both mobile and desktop, as well as ensuring that capabilities such as role validation (through manual attempts) worked as expected.

## 9. Conclusion, summary, and evaluation of the achievements

Find A Placement is a web application developed to encourage students to undertake placements within the industry and help remove barriers by matching them with potential employers. The important outcome was having a minimum viable product within a two-week period with core features implemented (e.g. match notifications and algorithmic matching) using a scrum framework.

Whilst challenges arose in terms of familiarity and rapid cycles with the scrum framework, the team quickly adapted by using the backlogs as a basis and sometimes working together to complete a task. This proved helpful for the work items with a higher scale and priority such as automated email alerts

Furthermore, the overall development of the project highlighted throughout the report displays its capabilities and its practicality as a tool for students, making it a reasonable alternative to LinkedIn and Indeed in principle.

## 10. References

## 11. Appendices

### Figure 1.1

Questions

1. Are there any other features you would like to see in the final product outside of the brief for example light and dark mode.
2. Should a user be logged in to view placements? Or someone can view the placement without login, they only login when they actually apply for it.
3. Should a user be able to manually set their skills or should it be automatically pulled from their CV
4. Should the employer and students be notified of potential matches every time they occur or in batches or per day
5. Would you like us to include accessibility options for users and employees (changes in colour scheme, font size etc) such as switch light and dark mode
6. What is the prefer database
7. What do you prefer front end? React , Plain JS, bootstrap or plain CSS, tailwind
8. What language do you prefer for back end (PHP, java, python, node.js)
9. Would you like use to create figma design before we start developing the actual website
10. Does the website only include placements that are a year long or are placements that are e.g. 3 month, 6 months long still acceptable

### Figure 1.2

## Requirements

By Abdur Sujon 6 min See views Add a reaction

### MVP (Minimum Viable Product) — FULL LIST

*Note: Each feature must be tested*

#### Summary of must feature

1. Admin
2. User Registration & Login
3. Student Profile
4. Employer Profile
5. Create Placement Listing
6. Basic Matching Engine
7. Display Matches
8. Basic Notifications

#### 1. Admin:

**First thing first (At least 8/9 Admin account) . We need someone to manage the platform who can do CRUD in the main System.**

Even in the simplest version, someone must be able to:

- Remove bad accounts
  - Fix incorrect placements
  - Moderate content
  - Reset passwords or unlock users
- Without admin access, the entire system becomes unmanageable.

#### 1.1 Admin Authentication & Access Control

- Admin login (separate from student/employer view)

## Figure 1.3

### Final MVP Feature Summary

1. Admin
2. User Registration & Login
3. Student Profile
4. Employer Profile
5. Create Placement Listing
6. Basic Matching Engine
7. Display Matches
8. Basic Notifications

1. As an Employer I want to provide contact details (contact name, email address, phone number and postal address).
2. As an Employer I want to provide placement details (placement description, skills required, salary, location, preferred start and end date)
3. As a website wants to provide information about the benefits of placements to both students and employers.
4. As a website wants match placements and students (dating agency esk) and contact employers and students when a match occurs
  - a. As an Employer I want to provide contact details (contact name, email address, phone number and postal address).
  - b. As an Employer I want to provide placement details (placement description, skills required, salary, location, preferred start and end date)
  - c. As a website wants to provide information about the benefits of placements to both students and employers.
  - d. As a website wants match placements and students (dating agency esk) and contact employers and students when a match occurs

## Figure 1.4

Need to defend the reason of using the technology

**<!-- Feature 1: Student accounts -->**

- Account ID (int)
  - should be hidden from the user
- First Name
  - cannot be considered a malicious word
- Last Name
  - cannot be considered a malicious word
- Email
  - must be real
  - could include verification
- Phone number
  - must be acceptable format
  - could include verification
- Postal address
  - Should be of an acceptable format
  - must be real (cannot be fake)
- Password hash
  - Cannot be seen by user, user should input password and be converted to hash
- CV URL
  - CV must be of a PDF format
  - Should be checked for malicious words/images
- Profile Picture URL
  - ensure that they are of an acceptable format
  - Should be checked for malicious imagery
- Placement applications
  - View active applications / previously applied to placements

**Figure 1.5**

## <!--Feature 2: Employer account -->

### Explicit features

- Placement Description
  - unclear information - must be to the point, no stories, no alternate skill names
  - false information - like inflated list of skills required, benefits
  - Vague work conditions
- Skills required
  - unrecognised skill - alternate names, miss spellings
  - Too general - wrong context, vague work/Job titles
- Salary offered
  - Formatting - force currency type (pound,dollar etc), no negative numbers, clear pay
- Location
  - must ensure location is sensible (e.g. nothing in the antarctic)
  - could ensure location is real (dictionary?,list?,maps?)
  - give option for hybrid,onsite,remote
  - should give option for multiple location
- Preferred start and end date
  - need to ensure that the start and end date are a year long (until told otherwise)
  - need to ensure acceptable dates are inputted (end date cannot be before start date, if allowing user to type in dates must ensure it is in correct format and give warning)
  - should be SQL injection proofed

### other potential features

- Profile picture
  - correct document type
  - could ensure nothing graphic
- Placements
  - allow employer to view all current placements and delete/end the placement recruitment in advance
  - should be able to edit a placement and upload the changes
- Deadline
  - placement should have an explicit deadline for when the applications will no longer be considered
- Company details
  - info like location, contact information
  - notification of changes
- Employer login information
  - email/username
  - password
    - should be stored as a hash
- Employer verification
  - An employer should be able to be verified as the actual company (limit potential scam openings for personal details)

**Figure 1.6**

## <!--Feature 3 browsing users -->

- Must login in to apply for a placement
- Can view placements and companies

**Figure 1.7**

| Project Section | Tool  | Justification                                                                                                                                                                                              |
|-----------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Font end        | Figma | Used to create the UI layout as it was easy to allow the team to access the same file for collaborative work, without needing to install another software as it was a browser design and prototyping tool. |

|            |            |                                                                                                                                                                                                                                                                                                      |
|------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | Canva      | Similar to Figma it was used to design the initial stages of the applications as well as dictate the direction of operations.                                                                                                                                                                        |
|            | Bootstrap  | The time constraint and with more important requirements that need to be added to the application, The CSS was mainly done with bootstrap as not only did the team understand how to use it, it allowed us to create the dynamic website as well.                                                    |
|            | CSS        | Mainly using bootstrap for most of the design, CSS was for creating a more personal look then the template look of bootstrap.                                                                                                                                                                        |
|            | JavaScript | Used to create a more personal curated website then the templates provided by bootstrap.                                                                                                                                                                                                             |
|            | HTML5      | It is the latest version for creating web sites that compatible with new browsers and the team is familiar with using the language.                                                                                                                                                                  |
| Backend    | PHP        | Chosen and used in PhpStorm editor, as the team was familiar with using both the editor and language to create HTML, CSS, JS.<br>The application was also download for a previous project. This means the teams time needed to download and setup, was used to work on research or initial features. |
|            | SQLite     | For creating are databases, the team knows how to create and insert data into this database, it allowed for quick and simplified creation of tables and data entries for the initial database.                                                                                                       |
| Deployment | Poseidon   | It was the server chosen by the university to host the application.                                                                                                                                                                                                                                  |
| Management | Jira       | This will be the main site in which information is shared, set timelines and distribute tasks.<br>Figure 2 shows the full workflow on Jira.                                                                                                                                                          |

|  |          |                                                                                                                                                |
|--|----------|------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Teams    | Designated as the main contact form with the clients on the progress of the code being both sides had the initial setup.                       |
|  | WhatsApp | Allowed the team to stay in contact and keep updates on each other's works.<br>The group chat was also set up before the brief was accessible. |

Figure 1.8 Management using Jira

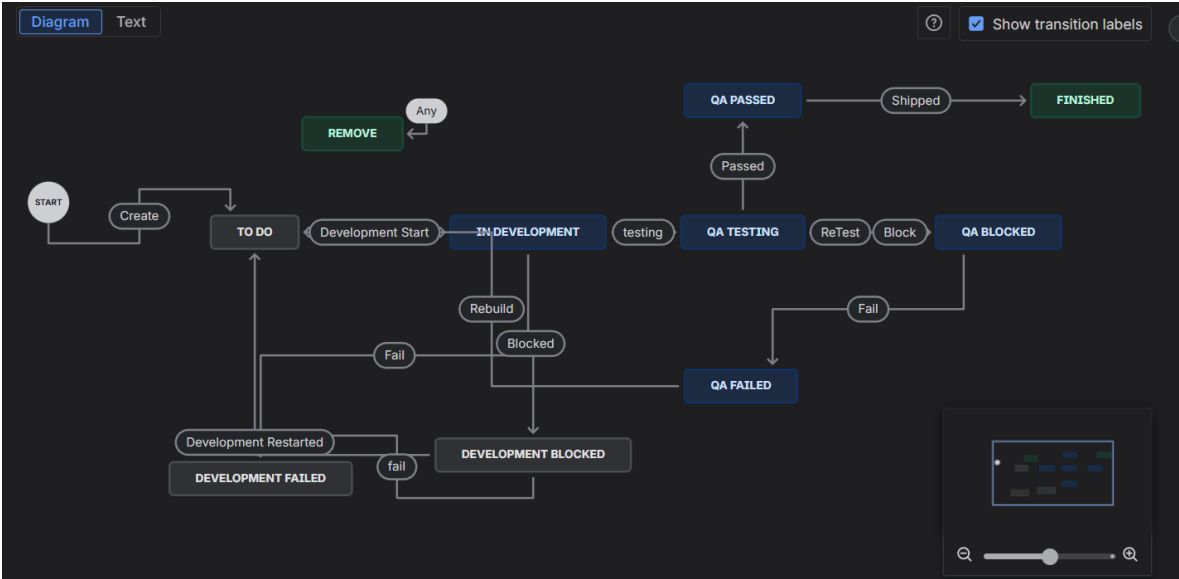


Figure 1.9

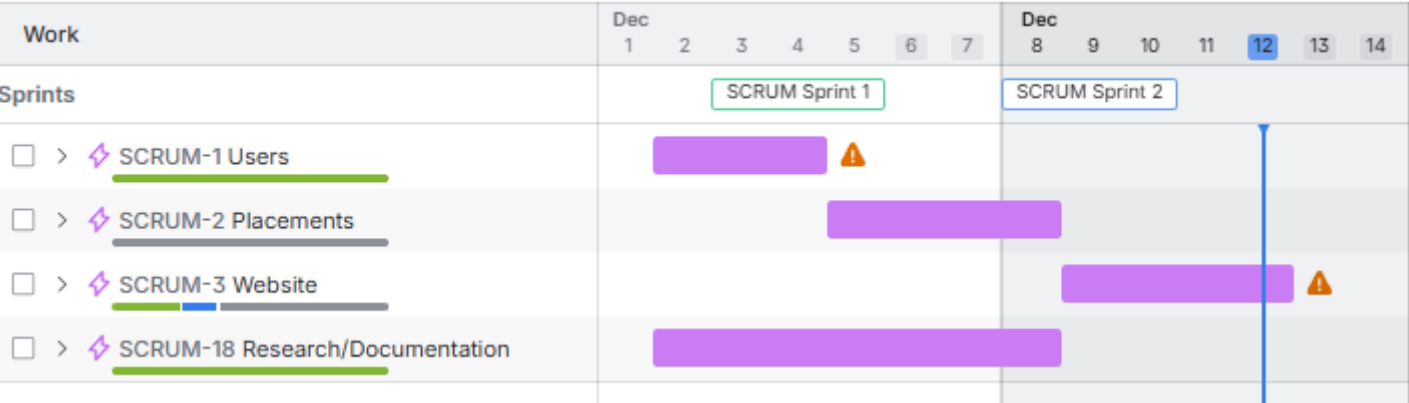


Figure 2

**1. Finalize the tool as team**

**2. Some clear boundaries:**

- a) We will work on pair for the assign tasks without disturbing each other work flow.
  - b) We will only communicate to the scrum master for clarity on tasks we are unsure or stuck, in under any circumstances we will try out best not to distract entire team members.
  - c) We will hold group meeting for team discussion, then we can communicate.
- Prevents chaos and mid-sprint disruption.

**3. Design:**

- a) Home page + employer + design + implementation (Jacob, Sujon)
- b) Article page + implantation (Jon, James)
- c) Admin page design + implementation (Henry, Joe)
- d) Implement SQL, and admin page with automated tests (Henry, Sujon, Joe)
- e) Implement home and article page ( Jacob, James, Jon)